1. Client Requirements & Scenario Overview

- **Project Title:** Ratlou Local Municipality Offices (Setlagole) Network Infrastructure
- **Student Details:** MZOBE, BAYA (Student Number: 44027516)
- **Client Profile:** Municipal Services / Local Government Office (Setlagole)
- **Scope of Work:** Design and deploy a segmented, secure, and scalable local area network for administrative staff, public wireless access, and future infrastructure expansion using an assigned /24 root addressing block.

2. Network Design & Architecture

- **Hierarchical Topology Model:** Core/Distribution layer handled by SW1-Core connected directly to edge access switches (SW2-Admin, SW3-Public, and SW4-Floor2-CR2).
- **Router-on-a-Stick Inter-VLAN Routing:** Router R1-Ratlou uses sub-interfaces over a single 802.1Q trunk link to facilitate inter-VLAN routing and enforce traffic segmentation.
- **Security & Isolation:** Administrative traffic (VLAN 10) is logically separated from public Wi-Fi traffic (VLAN 20) and management traffic (VLAN 99). Wireless access is restricted exclusively to public zones and authenticated expansion areas.

3. IP Addressing Scheme (VLSM Allocation Table)

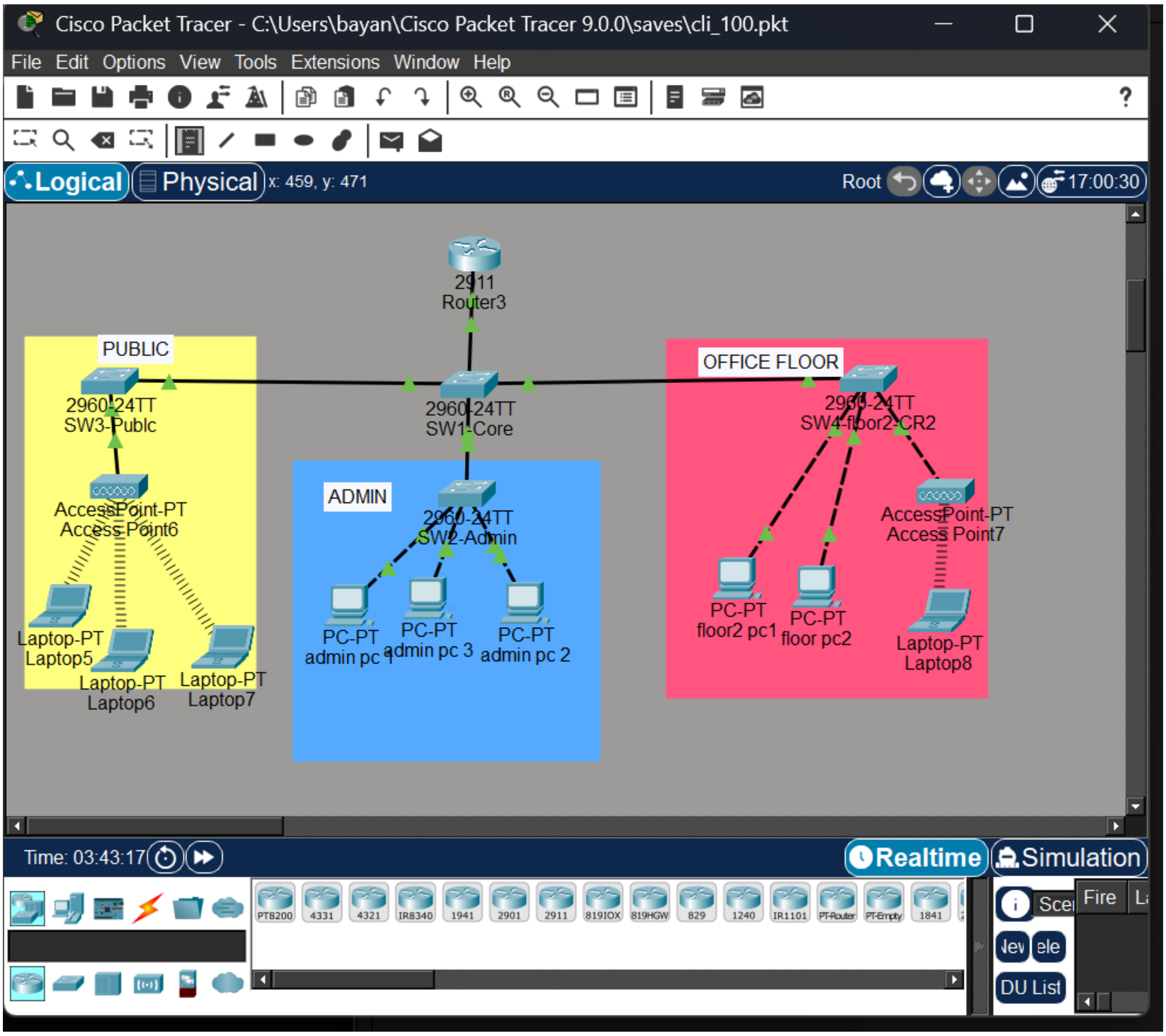
Root Network Block: 192.168.44.0/24

Subnet / Function	VLAN ID	Network Address	Subnet Mask	Usable Host Range	Default Gateway
Admin & Staff	VLAN 10	192.168.44.0/26	255.255.255.192	192.168.44.1 - 192.168.44.62	192.168.44.1
Public Wi-Fi Zone	VLAN 20	192.168.44.64/27	255.255.255.224	192.168.44.65 - 192.168.44.94	192.168.44.65

Subnet / Function	VLAN ID	Network Address	Subnet Mask	Usable Host Range	Default Gateway
CR2 Expansion Floor	VLAN 30	192.168.44.96/27	255.255.255.224	192.168.44.97 - 192.168.44.126	192.168.44.97
Management Subnet	VLAN 99	192.168.44.128/28	255.255.255.240	192.168.44.129 - 192.	192.168.44.129

4. Topology Layout

- **Diagram Overview:**



- **Device Inventory:**

- **Router:** R1-Ratlou (Cisco 2911 / ISR series)
- **Core Switch:** SW1-Core (Cisco Catalyst 2960)
- **Access Switches:** SW2-Admin, SW3-Public, SW4-Floor2-CR2 (Cisco Catalyst 2960)
- **Wireless Access Points:** Public AP (AP-PT) and Floor 2 WAP (AP-Floor2)
- **End Devices:** Admin PCs, Wireless Laptops, and Public PCs/Tablets

5. Packet Tracer Build Status

- Base network setup complete and functional.
- Inter-VLAN routing (802.1Q encapsulation) configured on router sub-interfaces.
- Active DHCP service running on R1-Ratlou for automated IP assignment across all subnets.
- Trunk ports configured across switches with native VLAN 99 alignment.

6. Configuration Commands (CLI Verification Logs)

code blocks pulled directly from show running-config in Packet Tracer:

1. Router (R1-Ratlou)

sub-interfaces, 802.1Q encapsulation, IP address assignments, and DHCP pool configurations:

```
! Router Sub-interfaces & Gateway Addresses
interface GigabitEthernet0/0.10
 description Admin-Staff-Gateway
 encapsulation dot1Q 10
 ip address 192.168.44.1 255.255.255.192
!
interface GigabitEthernet0/0.20
 description Public-Area-Gateway
 encapsulation dot1Q 20
 ip address 192.168.44.65 255.255.255.224
!
```

```
interface GigabitEthernet0/0.30
 description CR2-New-Floor-Gateway
 encapsulation dot1Q 30
 ip address 192.168.44.97 255.255.255.224
!
interface GigabitEthernet0/0.99
 description Management-Gateway
 encapsulation dot1Q 99 native
 ip address 192.168.44.129 255.255.255.240

! DHCP Pools Configuration
ip dhcp pool AdminPool
 network 192.168.44.0 255.255.255.192
 default-router 192.168.44.1
 dns-server 8.8.8.8
!
ip dhcp pool PublicPool
 network 192.168.44.64 255.255.255.224
 default-router 192.168.44.65
 dns-server 8.8.8.8
!
ip dhcp pool CR2Pool
 network 192.168.44.96 255.255.255.224
 default-router 192.168.44.97
 dns-server 8.8.8.8
```

2. Core Switch (SW1-Core)

Show VLAN creations and all trunk interface settings:

```
! VLAN Database
vlan 10,20,30,99

! Router Trunk Link
interface GigabitEthernet0/1
 switchport mode trunk
 switchport trunk native vlan 99

! Access Switch Trunk Links(for public, admin, and new office floor)
interface range FastEthernet0/1 - 3
 switchport mode trunk
 switchport trunk native vlan 99
```

3. Admin Switch (SW2-Admin)

Show uplink trunk port and access ports assigned to VLAN 10:

```
! Uplink Trunk to Core Switch
interface FastEthernet0/24
 switchport mode trunk
 switchport trunk native vlan 99

! Admin End-Device(3 PC's for now) Access Ports
interface range FastEthernet0/1 - 3
 switchport mode access
 switchport access vlan 10
```

4. Public Switch (SW3-Public)

Show uplink trunk port and access port assigned to Public Wi-Fi (VLAN 20):

```
! Uplink Trunk to Core Switch
interface FastEthernet0/24
 switchport mode trunk
 switchport trunk native vlan 99

! Wireless Access Point Port
interface FastEthernet0/1
 switchport mode access
 switchport access vlan 20
```

5. Floor 2 Switch (SW4-Floor2-CR2)

Show uplink trunk port and access ports assigned to CR2 Floor expansion (VLAN 30):

```
! Uplink Trunk to Core Switch
interface FastEthernet0/24
 switchport mode trunk
 switchport trunk native vlan 99

! Floor 2 Wired & Wireless Access Ports
interface range FastEthernet0/1 - 2
 switchport mode access
 switchport access vlan 30
!
interface FastEthernet0/10
 switchport mode access
 switchport access vlan 30
```

7. Testing & Verification

Include evidence of successful communication across subnets:

- **DHCP Verification:** Output from Show ip dhcp binding on R1-Ratlou.

```
Router3
Physical   Config   CLI   Attributes

use. Delivery of Cisco cryptographic products does not imply
third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.

A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html

If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco CISCO2911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
3 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!


%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.30, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.99, changed state to up


R1-Ratlou>enable
R1-Ratlou#show ip dhcp binding
IP address       Client-ID/              Lease expiration        Type
                 Hardware address
192.168.44.7     00D0.FF91.D3C5           --                     Automatic
192.168.44.6     00E0.F77B.11C0           --                     Automatic
192.168.44.72    0030.A3ED.7513           --                     Automatic
192.168.44.73    0001.43E8.8631           --                     Automatic
192.168.44.71    000C.CF22.07AC           --                     Automatic
192.168.44.75    00E0.8FB1.5294           --                     Automatic
192.168.44.104   0006.2A00.6188           --                     Automatic
192.168.44.103   00E0.8FB1.5294           --                     Automatic
R1-Ratlou#

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Top
```

- **VLAN Database Verification:** Output from show vlan brief on SW1-Core.

```
SW1-Core
Physical   Config   CLI   Attributes

SW1-Core con0 is now available

Press RETURN to get started.

SW1-Core>enable
SW1-Core#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24, Gig0/2
10   Admin-Staff                      active
20   Public-Area                      active
30   CR2-New-Floor                    active
99   Management                       active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SW1-Core#

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```
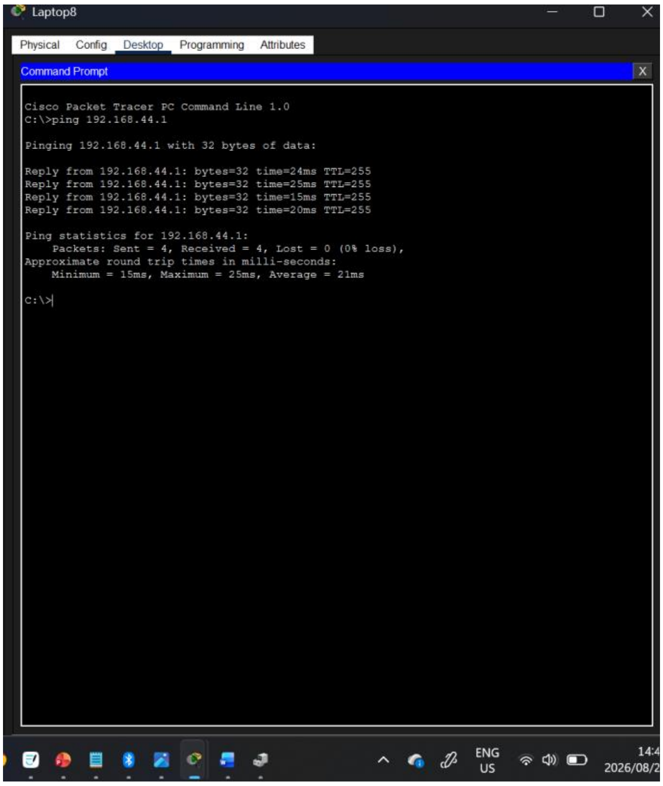

- **Trunk Link Verification:** Output from show interfaces trunk.

- **End-to-End Ping Verification:** Successful ping output from Floor 2 Wireless Laptop to Gateway 192.168.44.1:



8. Future Milestones Plan

- **Milestone 2 (02 October 2026):** Full implementation of Change Request 2 (CR2 expansion area) and complete Fault Isolation Scenario (controlled network breakdown, diagnostic logs, and resolution proof).
- **Final Submission (16 October 2026):** Final tested .pkt file, complete GitHub repository documentation, and 15–20 minute screen-recorded video presentation featuring webcam overlay and live connectivity demonstration.

9. Preliminary Project Reflection

Developing the foundational network for the Ratlou Local Municipality highlighted the importance of structured VLSM planning before applying CLI configurations. Managing native VLAN matching across trunk links initially triggered Spanning Tree protocol alerts, but aligning native VLAN 99 across all switches resolved the issue. The configuration successfully achieves inter-VLAN routing while isolating public Wi-Fi traffic from administrative databases.